

Universidad Nacional
Federico Villarreal

Facultad de Medicina

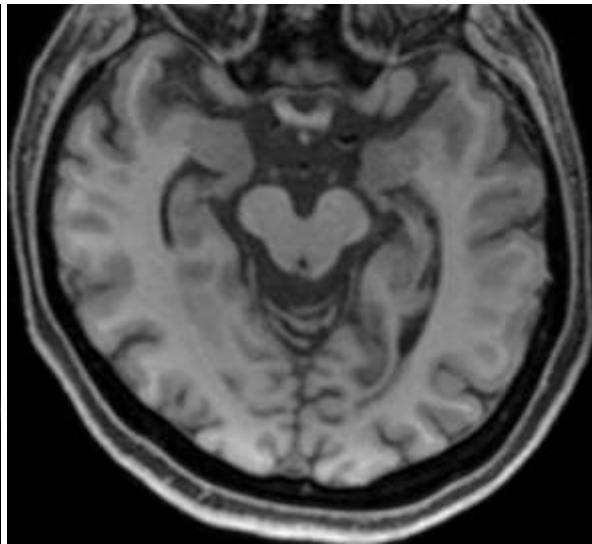
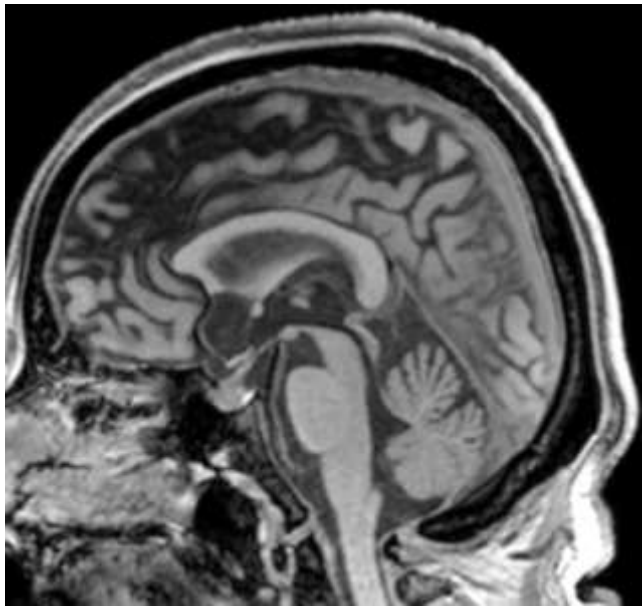
NEUROIMAGEN EN ENFERMEDAD CEREBROVASCULAR

Asignatura: NEUROLOGIA
Dr. Esteban Arias

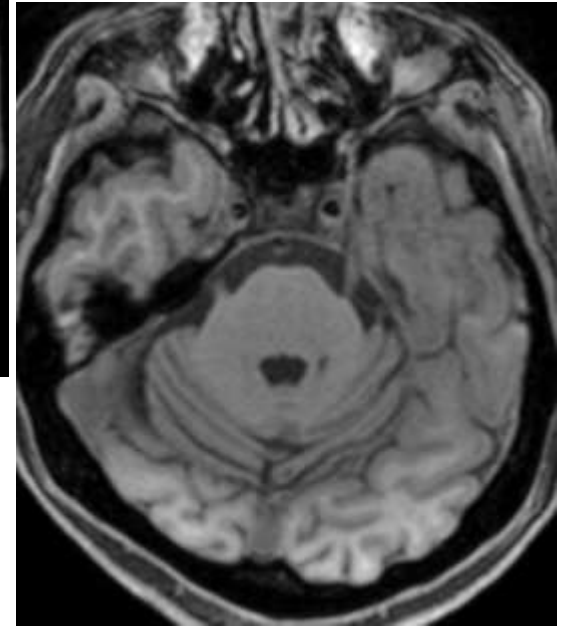
Semestre Académico 2026-II

METODOS DE ESTUDIO MEDIANTE IMAGENES DEL SISTEMA NERVIOSO

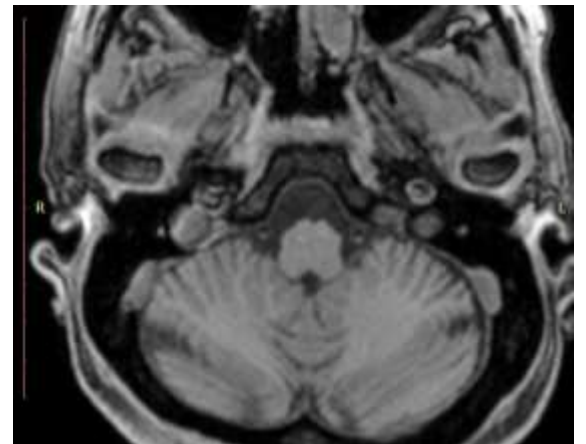
- **TOMOGRAFIA COMPUTARIZADA (TC):**
 - SIN CONTRASTE
 - CON CONTRASTE
- **ESTUDIO DE VASOS SUPRAORTICOS Y CEREBRALES MEDIANTE:**
 - RESONANCIA MAGNETICA (RM) O ANGIORESONANCIA
 - ANGIOTEM (angiografía usando Tomógrafo Espiral Multicorte)
- **RESONANCIA MAGNETICA (RM)**
 - SECUENCIA T1
 - SECUENCIA T2
 - FLAIR
 - DIFUSION
 - ADC
 - PERFUSION
 - ESPECTROSCOPIA
 - T2-FFE (ecogradiente)
 - RM- FUNCIONAL
 - VenBold



Mesencéfalo



Protuberancia

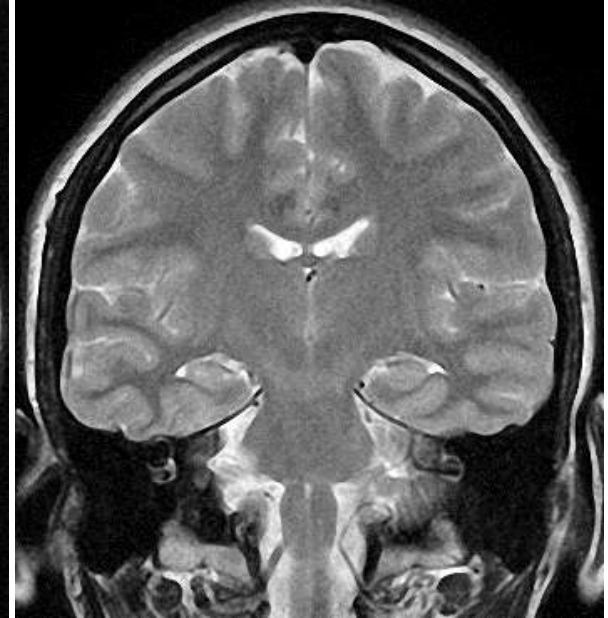
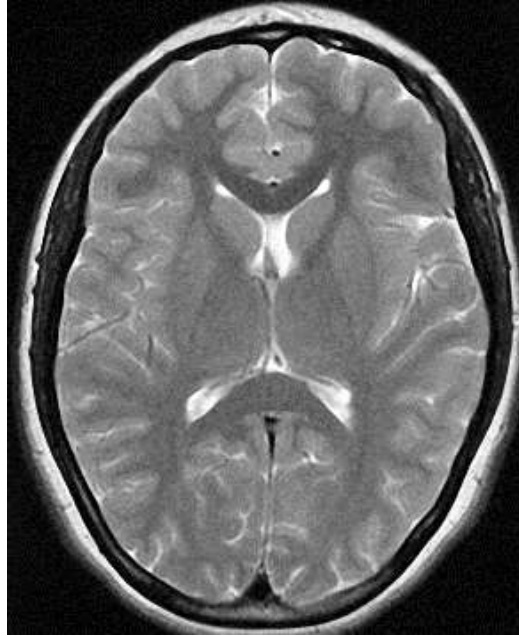
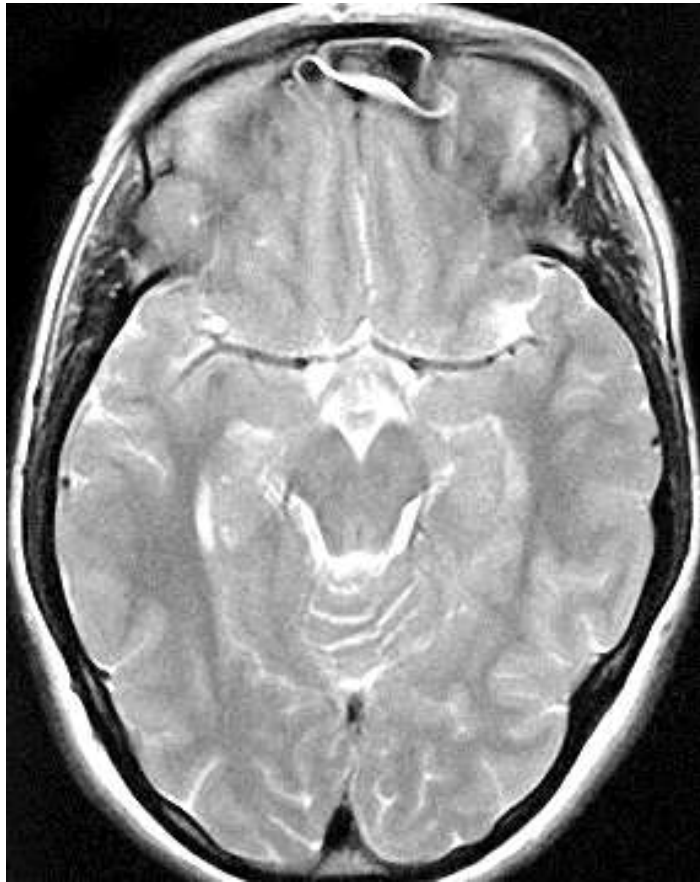


Bulbo



ANATOMIA NORMAL MEDIANTE RMN -T1:
proporciona detalles morfológicos.

RMN-T2: sensible a la presencia de agua. LCR: “color blanco” - “brilla”



EN RMN –T2:

- **ES HIPERINTENSO**

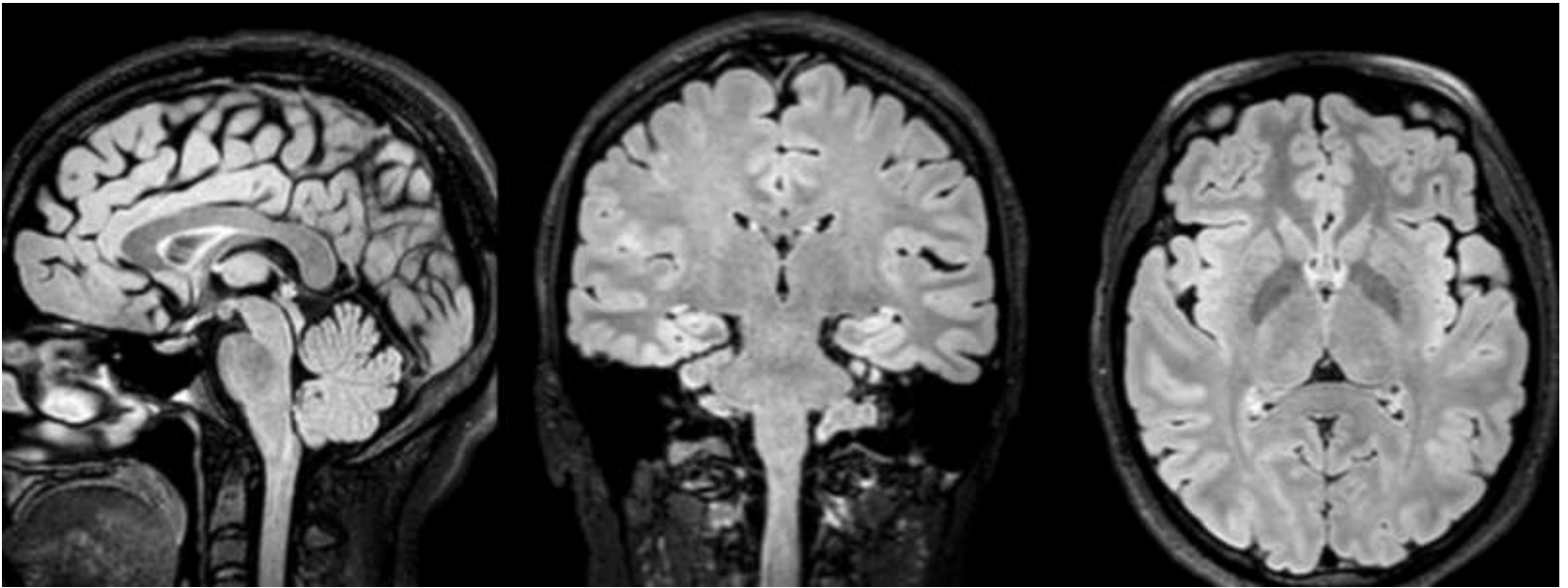
- LCR
- Quistes.
- Necrosis.

EN RMN –T2:

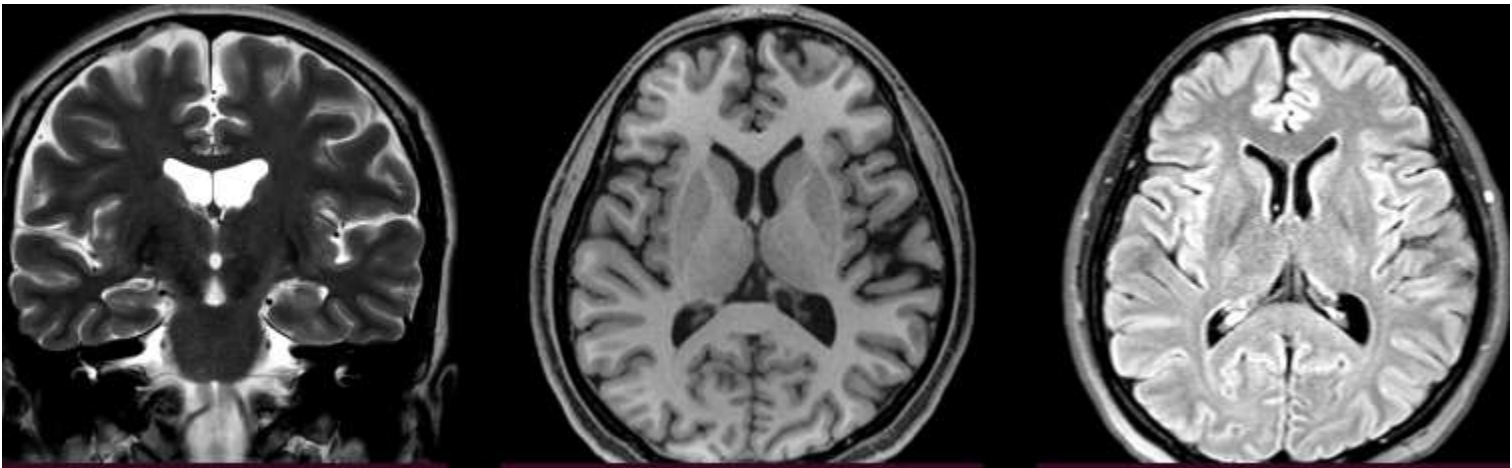
- **ES HIPOINTENSO**

- Vaso sanguíneo con flujo rápido

RM-Flair (Fluid Attenuated inversión Recovery)



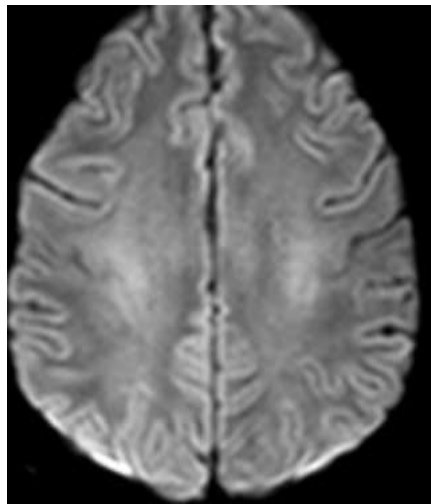
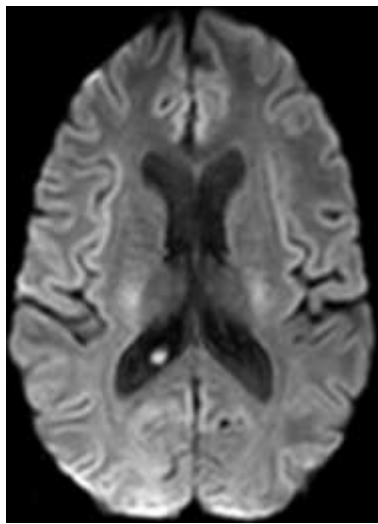
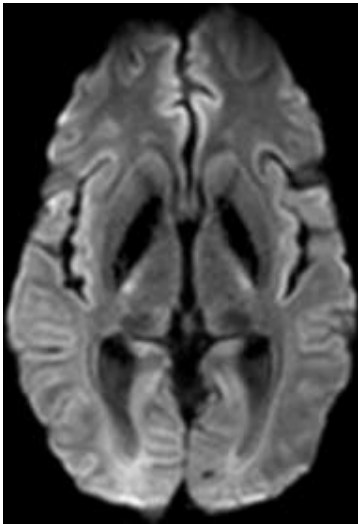
Esta secuencia elimina la señal del LCR, pero no la señal proveniente de lesiones patológicas que suelen presentar aumento en su contenido de agua. Por ejemplo, **edema**



RM-T2

RM-T1

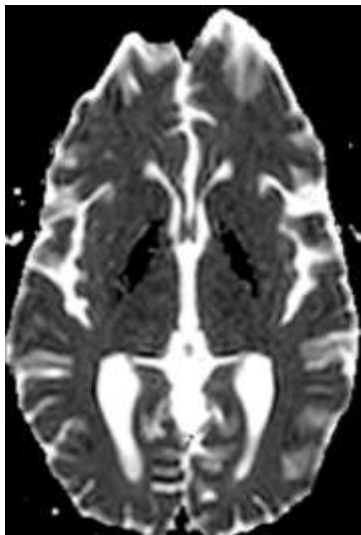
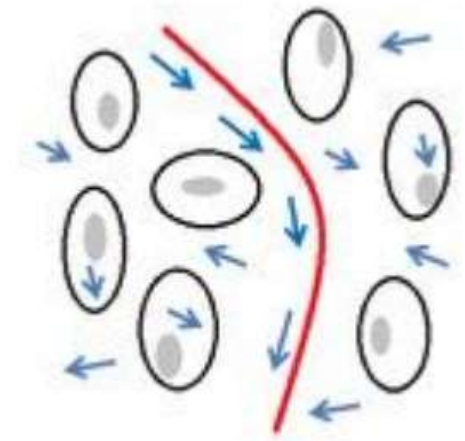
RM-Flair



RM-DIFUSION

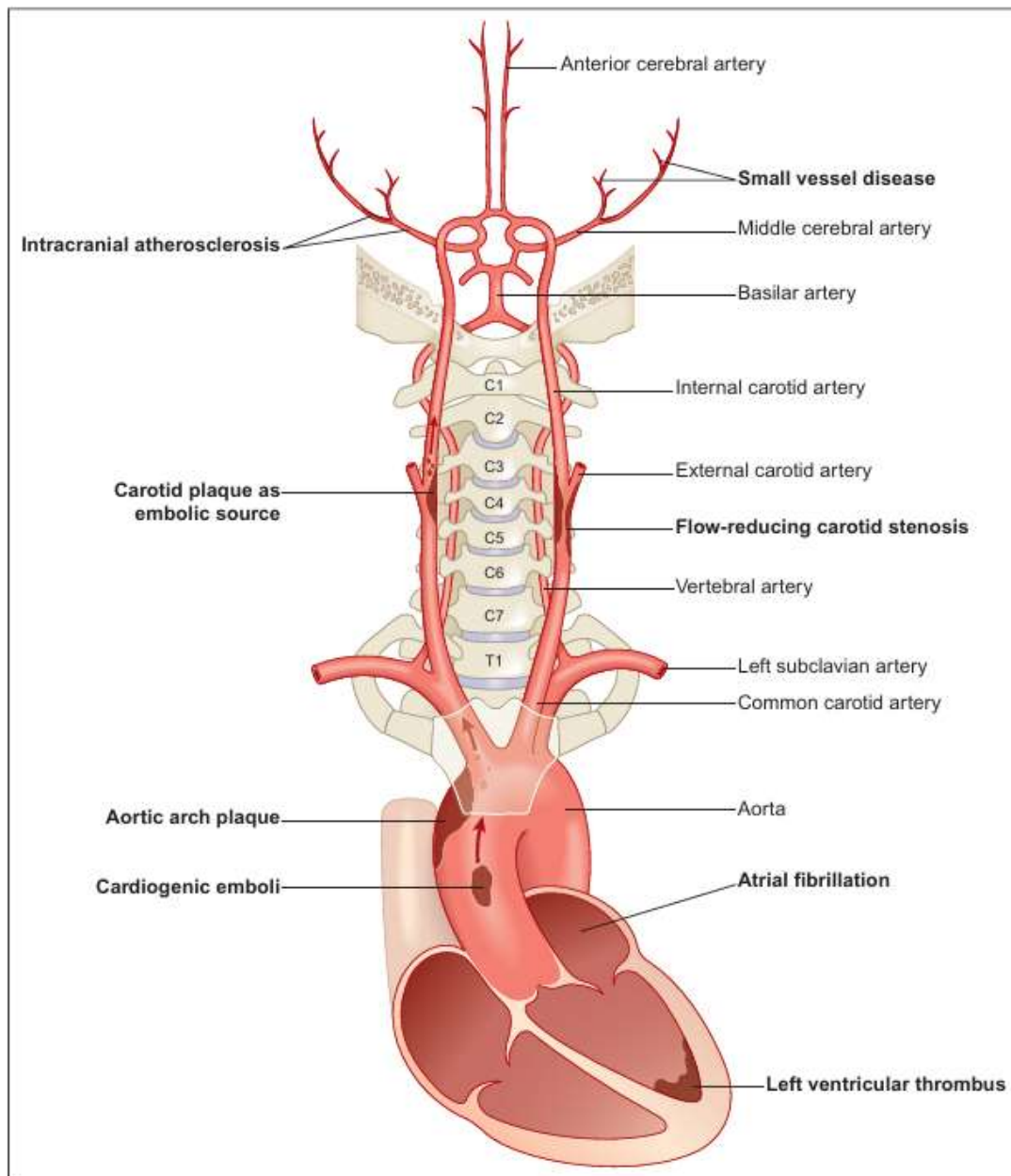
Representa la **difusión de moléculas** de agua en los tejidos biológicos.

Las moléculas de agua (flechas azules), se mueven en el espacio **intracelular, extracelular e intravascular** (microcirculación) de forma aleatoria (**difusión libre**)



RM-ADC (coeficiente de difusión aparente)

Permite **cuantificar** el grado del movimiento del agua o sea LA DIFUSION



METODOS IMAGENOLOGICOS PARA EVALUAR VASOS CEREBRALES Y SUPRAAORTICOS:



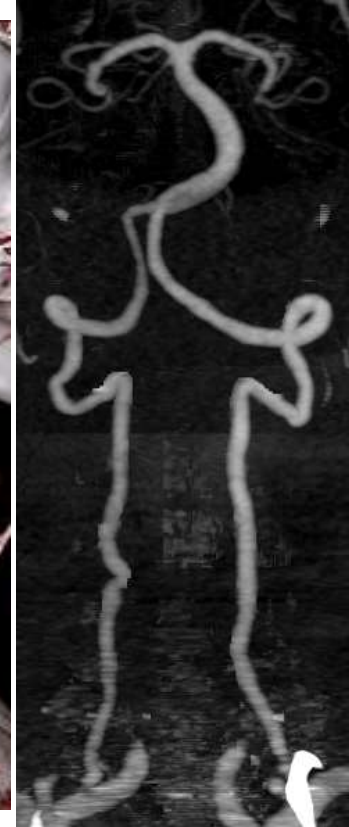
Angio resonancia: **sin y con contraste**



AngioTEM

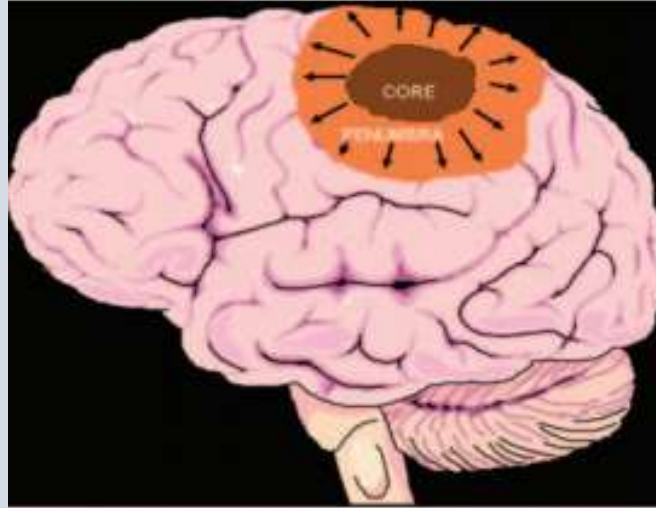


AngioRM



AngioTEM

ENFERMEDAD CEREBRO VASCULAR ISQUEMICA

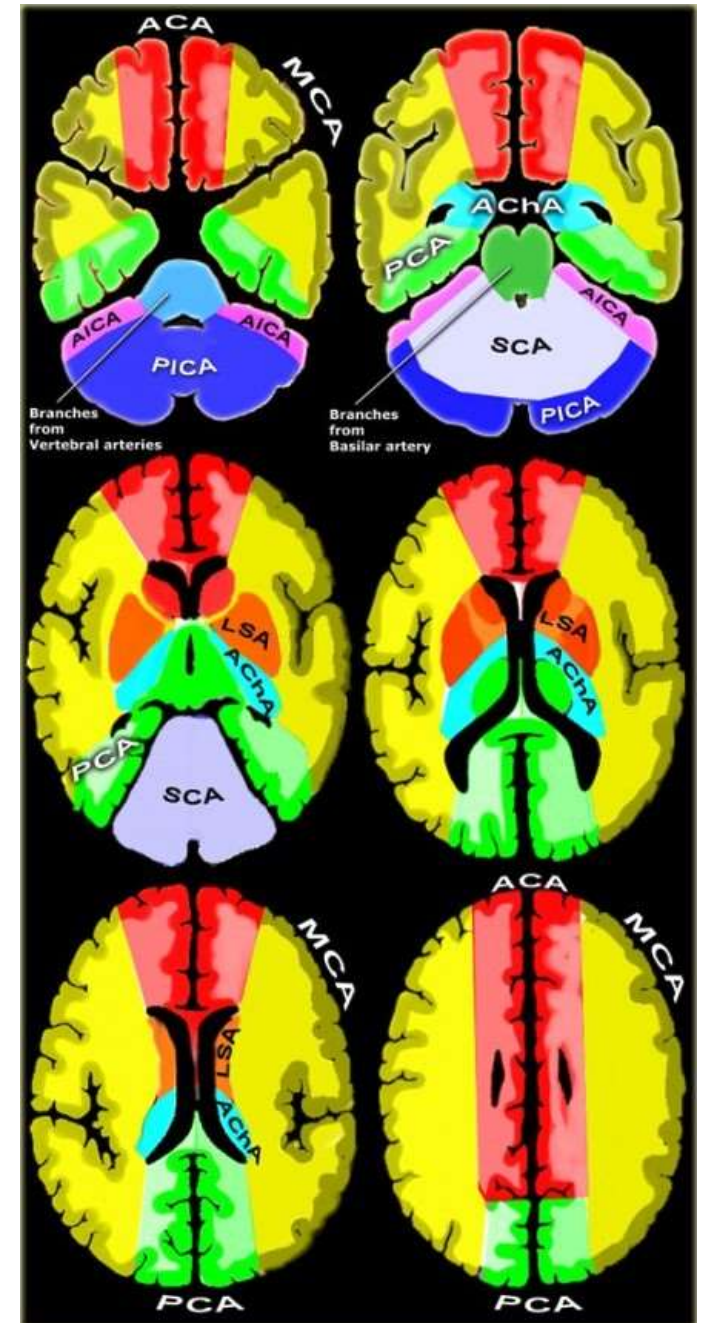


TOMOGRAFIA COMPUTARIZADA

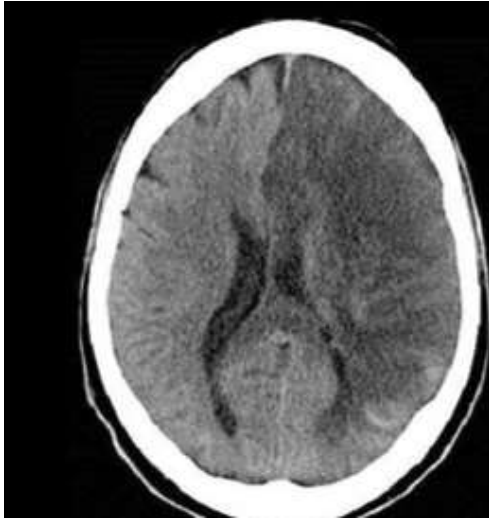
- TC sin contraste

RESONANCIA MAGNETICA

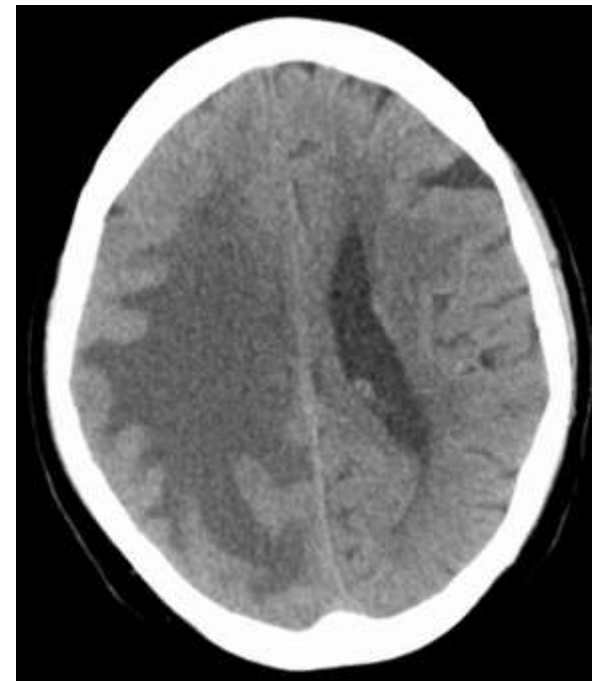
- RM -T1 y RM-T2
- RM-Flair
- RM-Difusion (DWI: Diffusión, Weighted Image)
- RM-ADC
- Angiografía por RM



EDEMA CEREBRAL



Edema citotóxico: con efecto de masa
Falla en el funcionamiento de la
bomba sodio-potasio (Na/K)



Edema vasogénico: con efecto de masa
Se altera la integridad de la barrera
hematoencefálica. Asociado a proceso
infeccioso, tumorales etc.

Paciente varón de 83 años

Con antecedente de HTA , con hemiparesia derecha



4 días después





TC: isquemia a las
6 horas

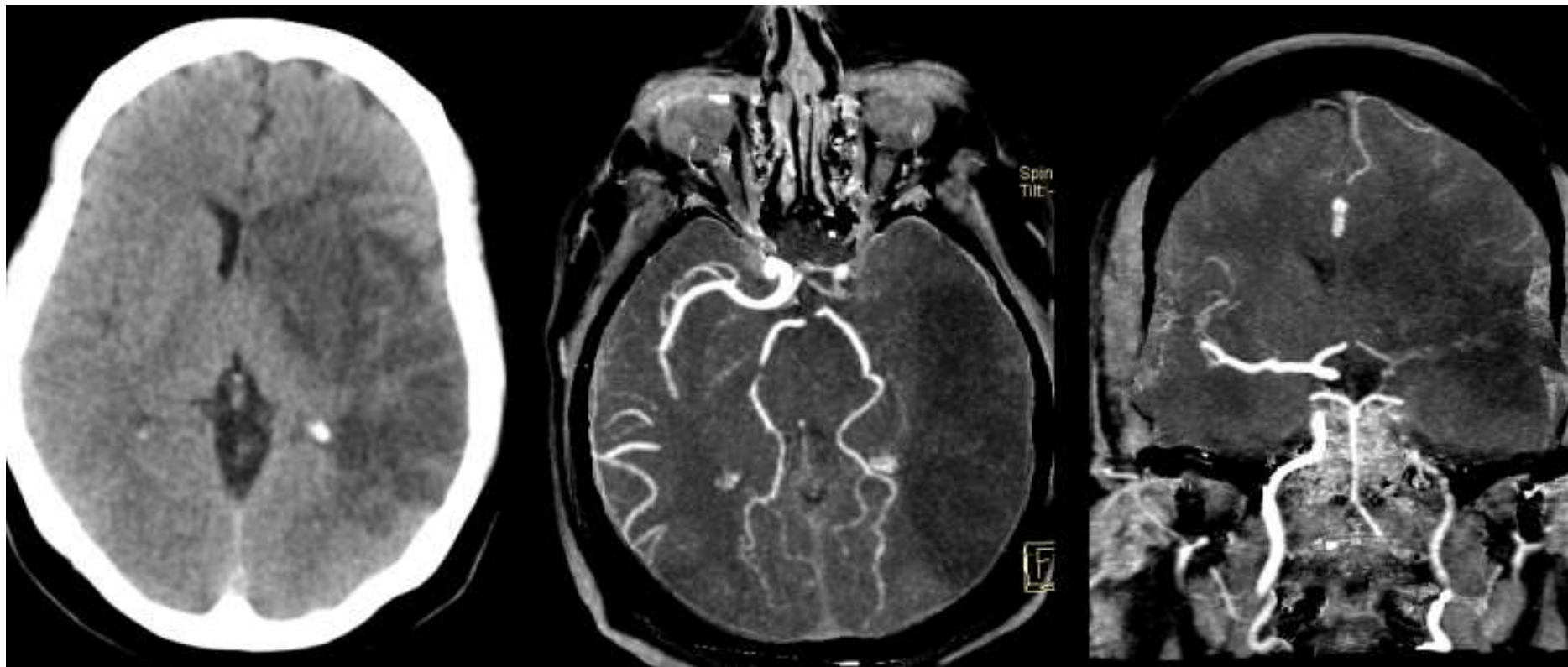
TC: isquemia a las
24 horas

TC: isquemia a las
40 horas

[Ischemic Stroke book.pdf](#)

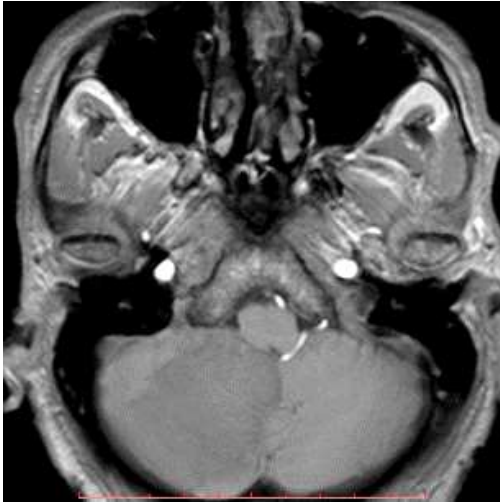


Signo hiperdenso de ACM

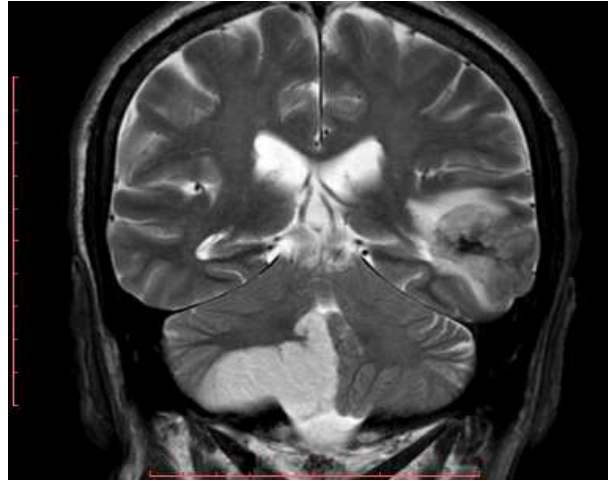


**ISQUEMIA EN EL TERRITORIO IRRIGADO POR ACM
MEDINTE TC**

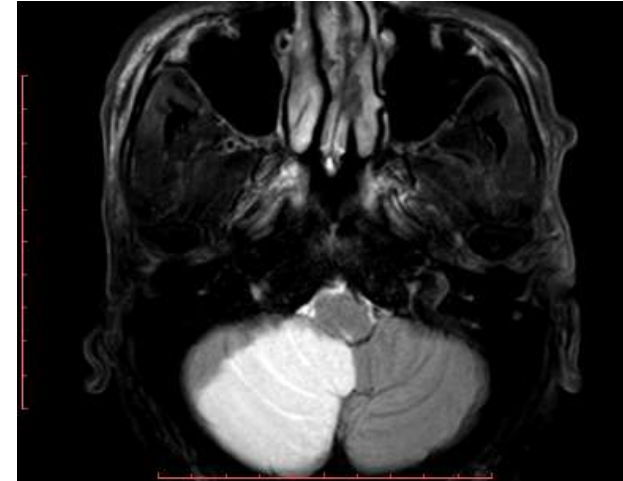
RM: ENFERMEDAD CEREBRO VASCULAR ISQUEMICA: agudo



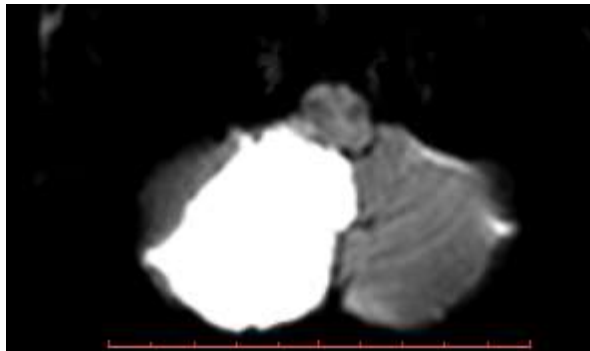
RM-T1



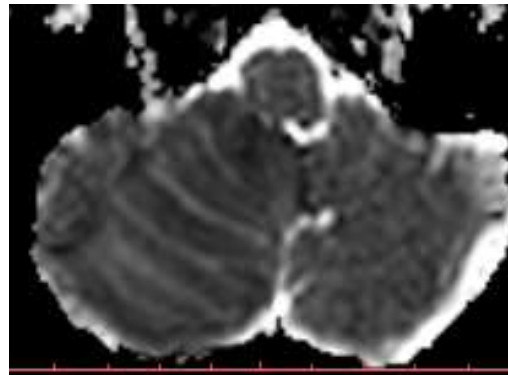
RM-T2



RM-FLAIR



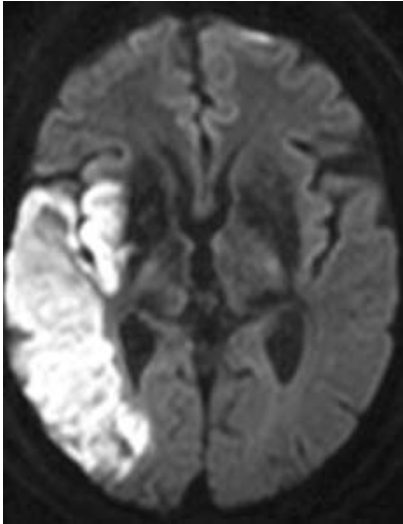
RM-DIFUSION



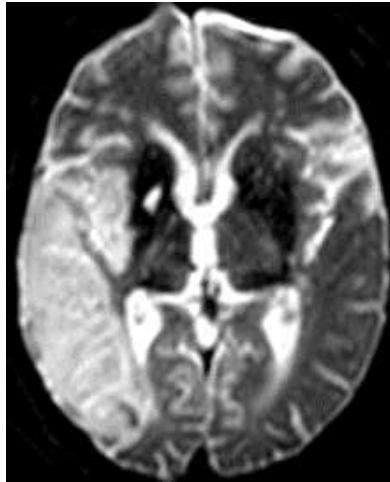
RM-ADC

RESTRICCION A LA DIFUSION se puede observar a partir de los 30 minutos de inicio de la ECV isquémica

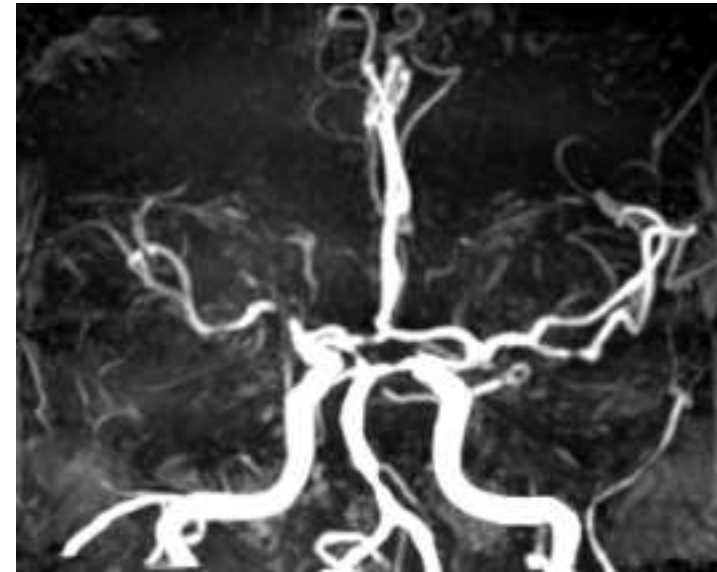
**RM: ENFERMEDAD CEREBRO VASCULAR ISQUEMICA:
subagudo**



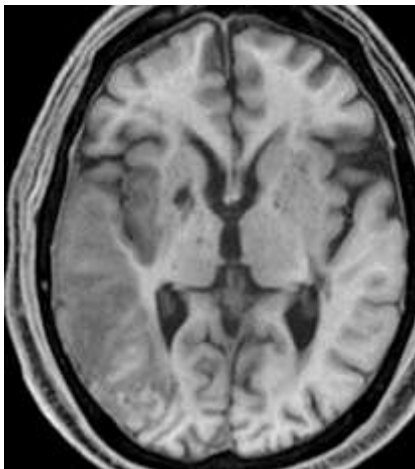
**RM-DIFUSION: RESTRICCION
A LA DIFUSION**



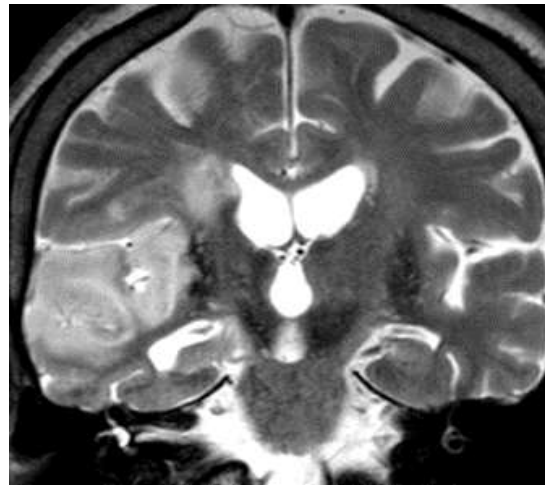
RM-ADC



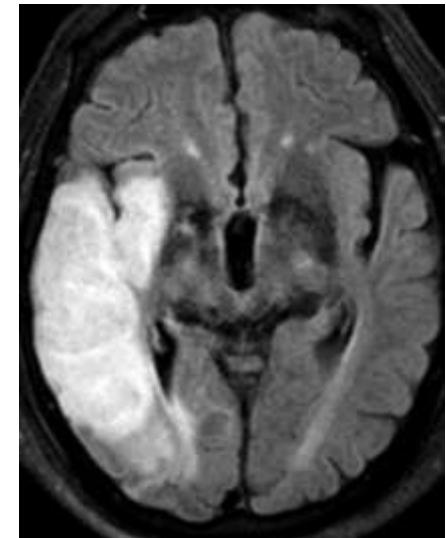
ANGIO-RM



RM-T1



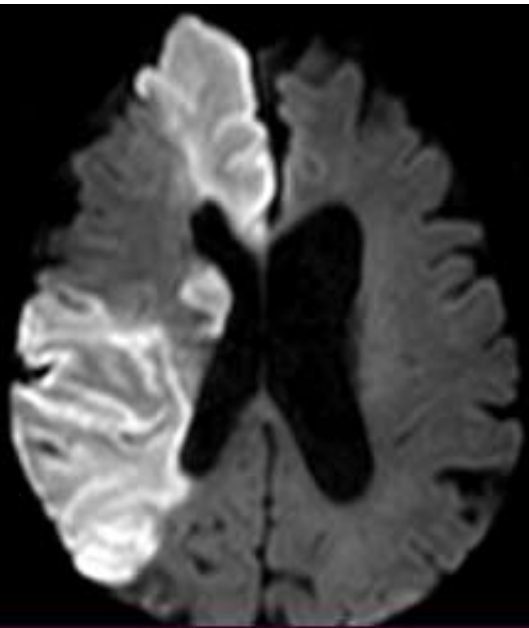
RM-T2



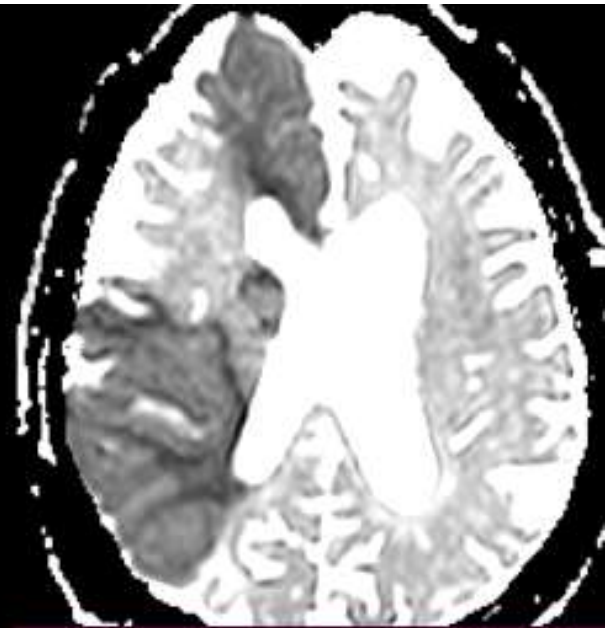
RM-FLAIR



TC



RM: DIFUSION **Restricción
a la difusión**

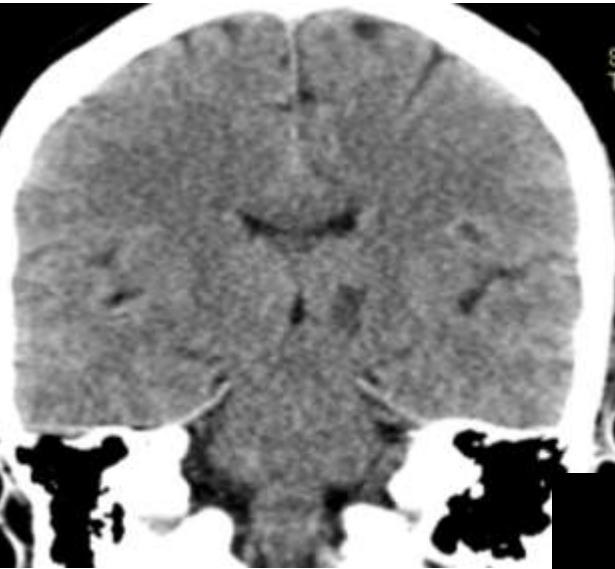


RM: ADC



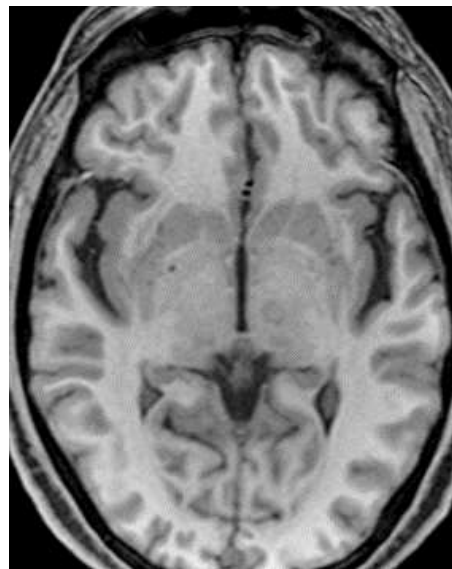
Angio-RM:
Estenosis de
carótida interna

ENFERMEDAD CEREBRO VASCULAR
ISQUEMICA: subagudo

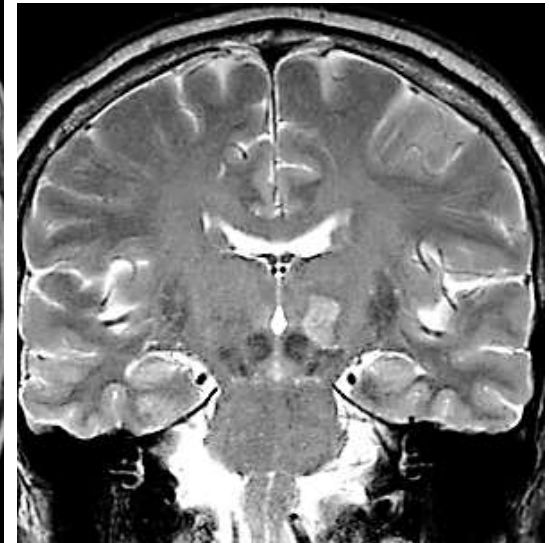


ISQUEMIA
EN tálamo

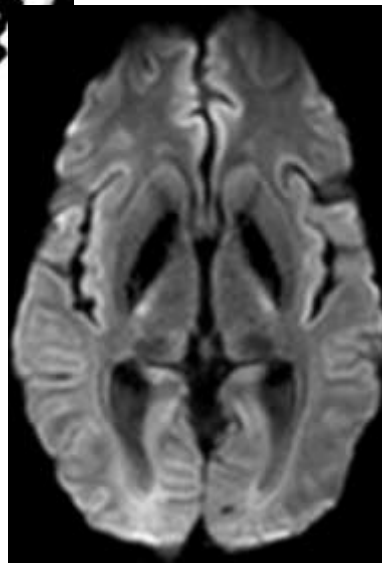
30 minutos
→



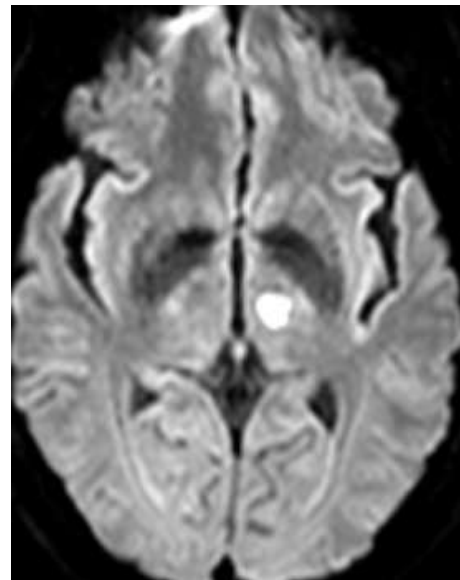
RM-T1



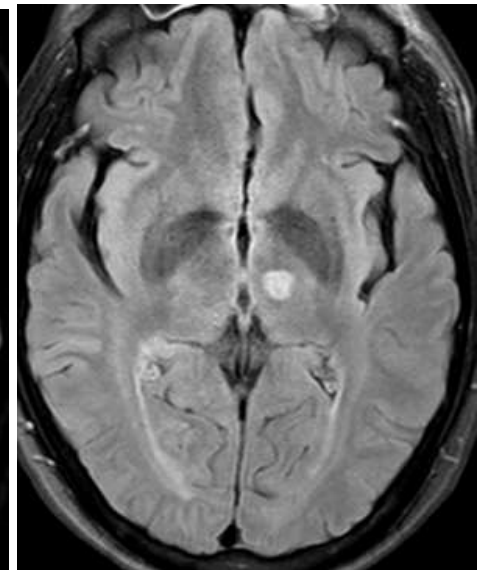
RM-T2



DIFUSION NORMAL



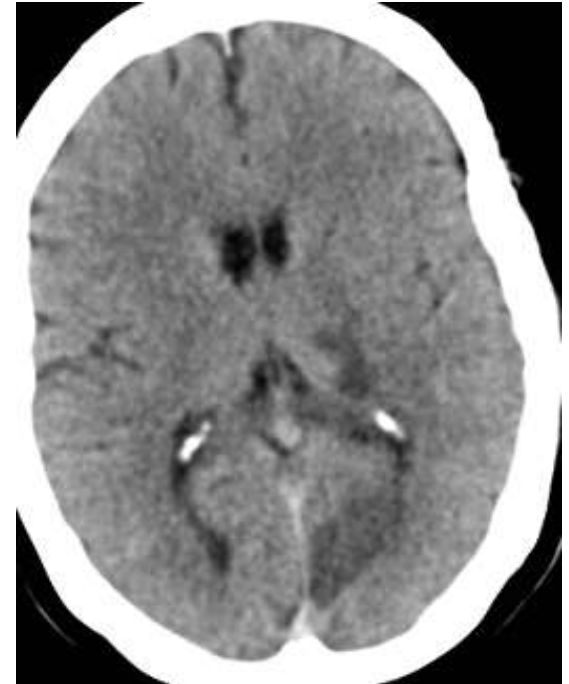
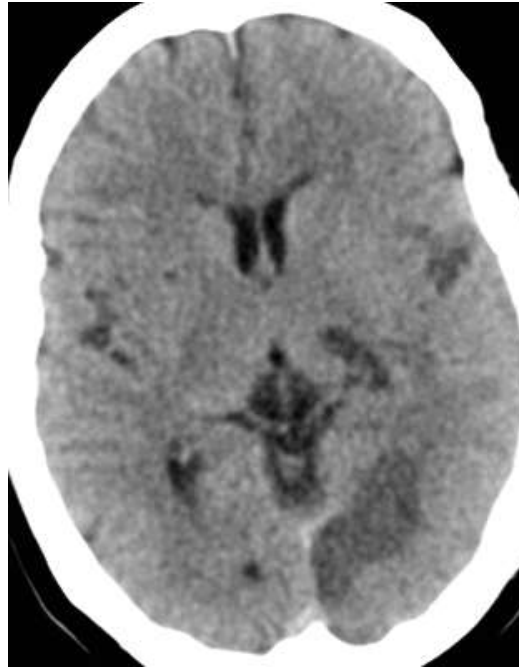
RM-DIFUSION:
RESTRICCION A DIFUSION



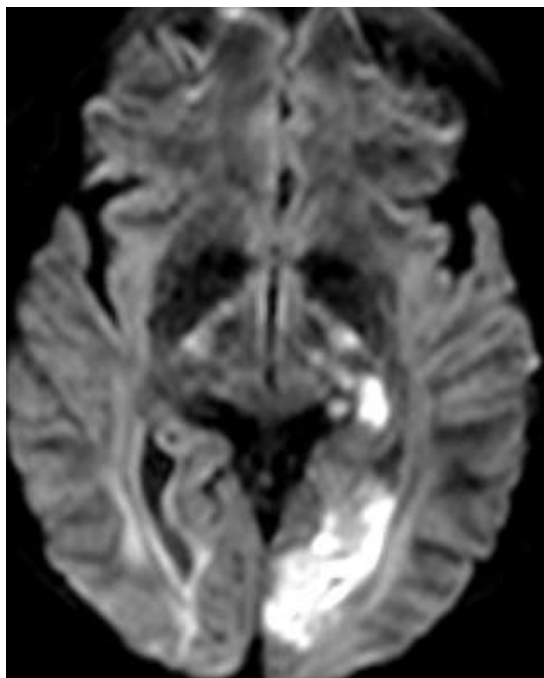
RM-FLAIR

Paciente mujer de 66 años antecedente de HTA, refiere que desde hace 2 días tiene problemas con la visión

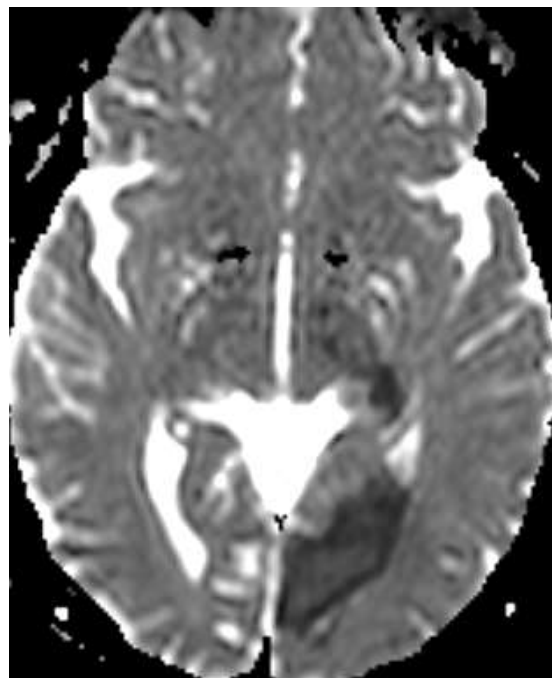
TEM realizada el 08-03 a 14.30 horas



RM REALIZADO EL 08-03 A 18.30 HORAS



RM-DIFUSIÓN

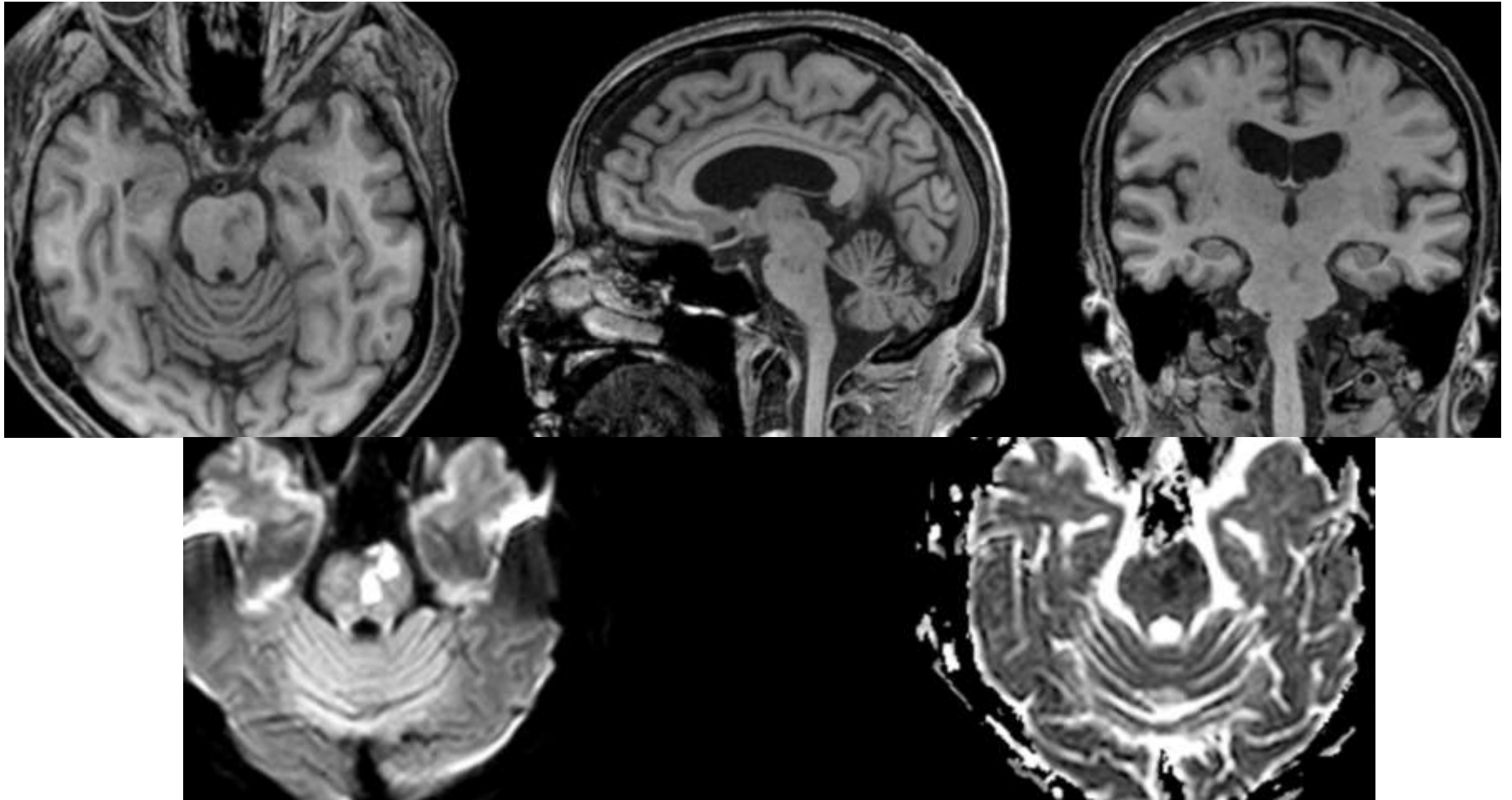


RM-ADC

ANGIORESONANCIA



TROMBOSIS DE ACP



RM-DIFUSION Isquemia en protuberancia

ADC: Isquemia en protuberancia



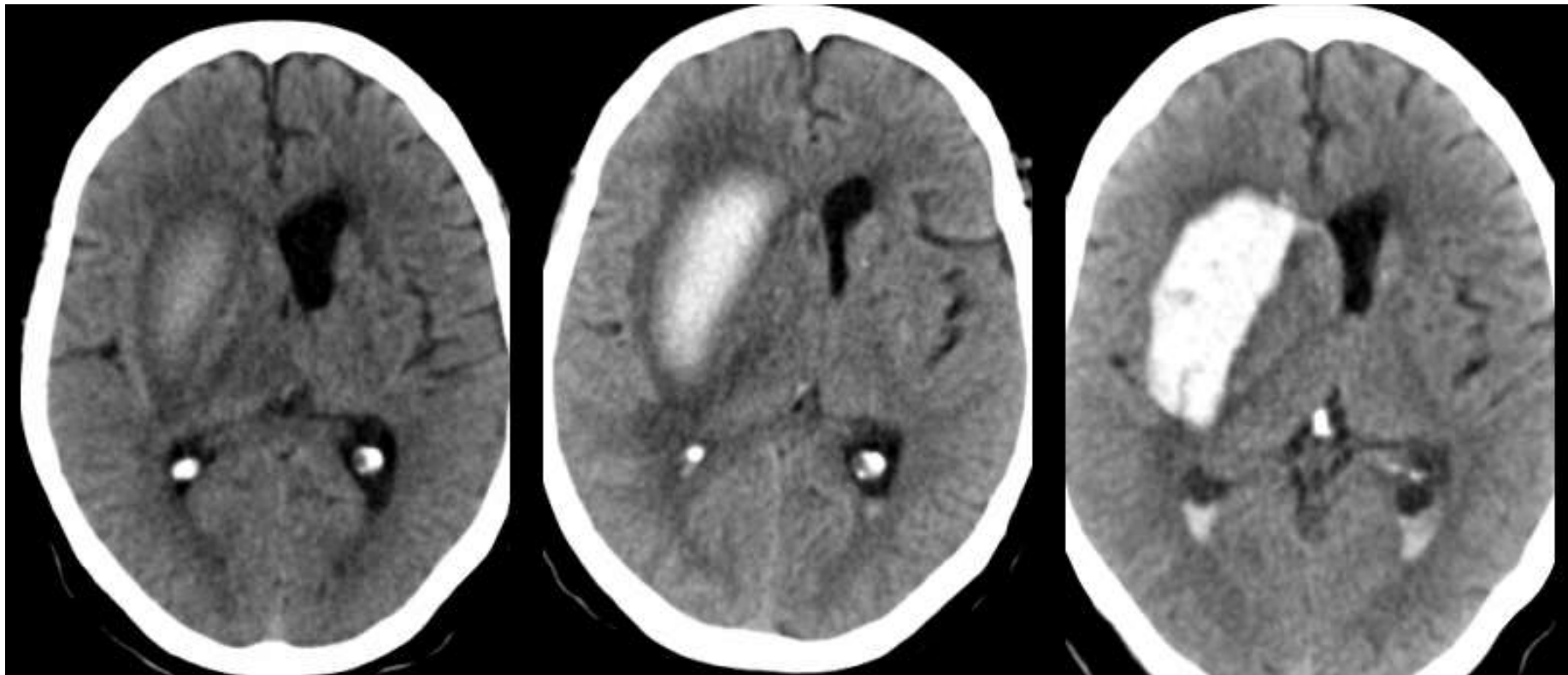
OCLUSION DE ARTERIA CAROTIDA INTERNA

VASOS SUPRA AORTICOS NORMALES



ESTENOSIS DE ARTERIA CAROTIDA INTERNA

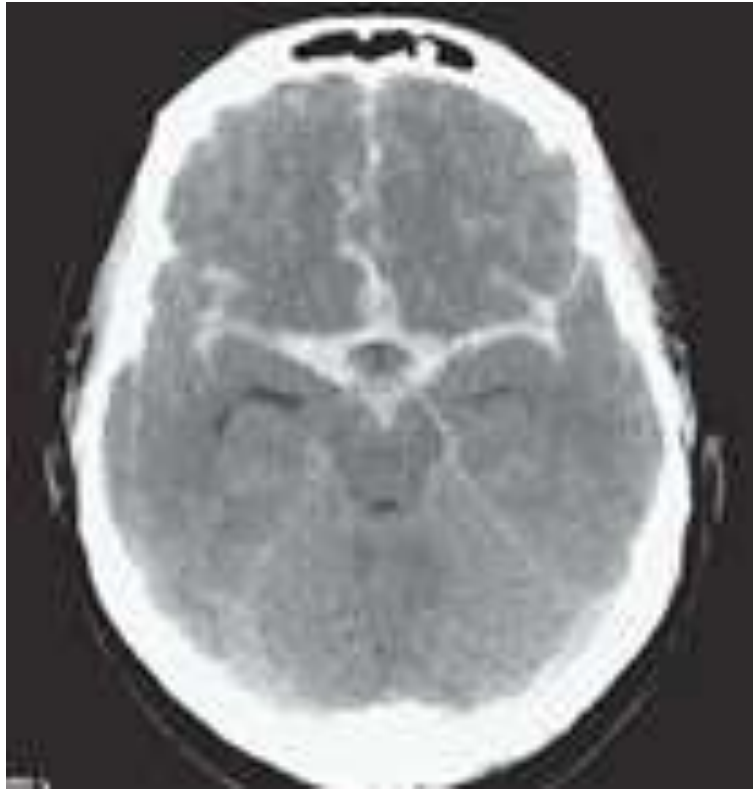
EVOLUCION DE LA HEMORRAGIA EN TC



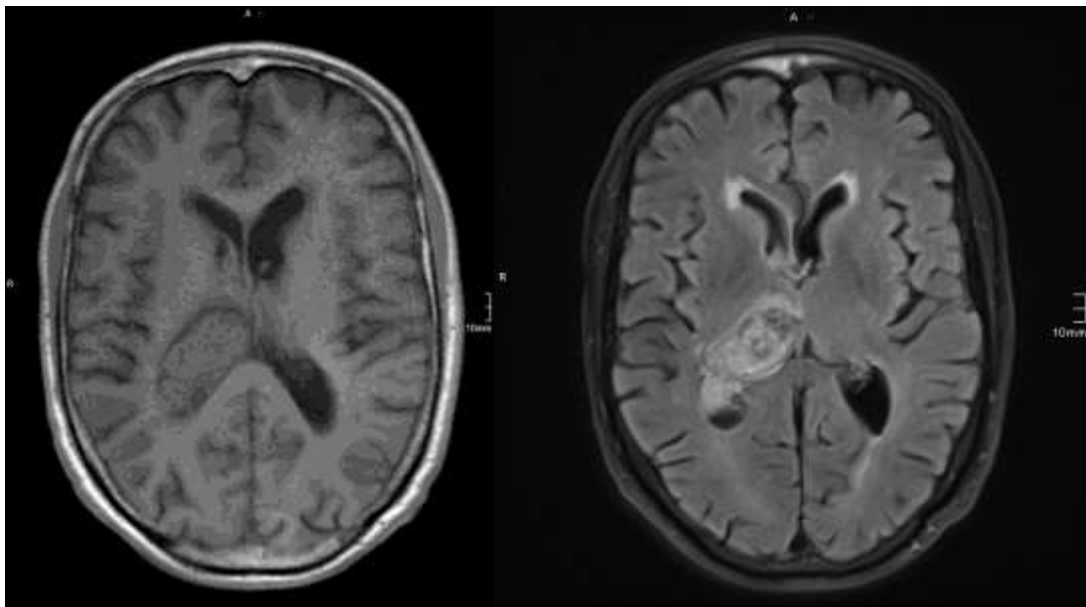
TC ULTIMO CONTROL 21-04

29-03

TC INICIAL 10-03



HEMORRAGIA SUBARACNOIDEA



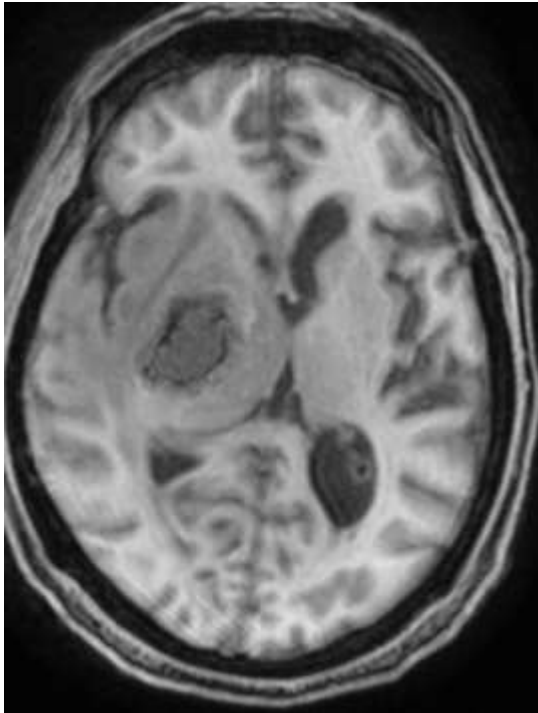
RM-T1

RM-Flair

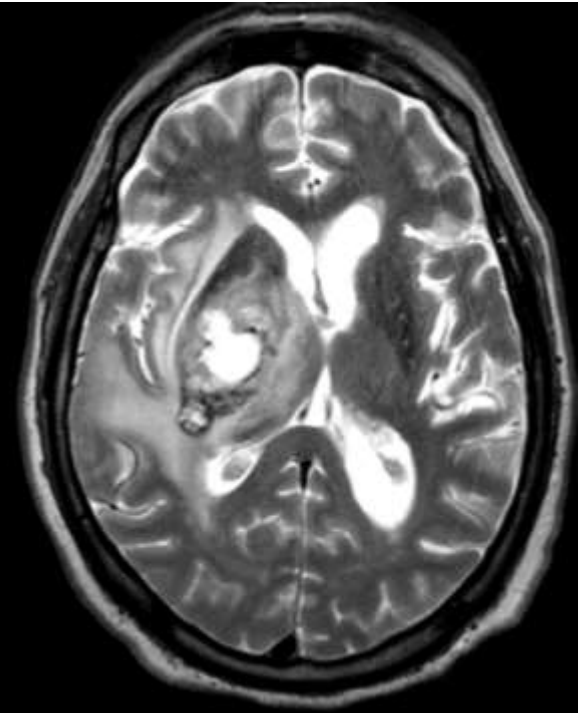
Hemorragia
hiperaguda menos
de 24 horas

<https://radiopaedia.org/articles/haemorrhage-on-mri-1>

RM-T1

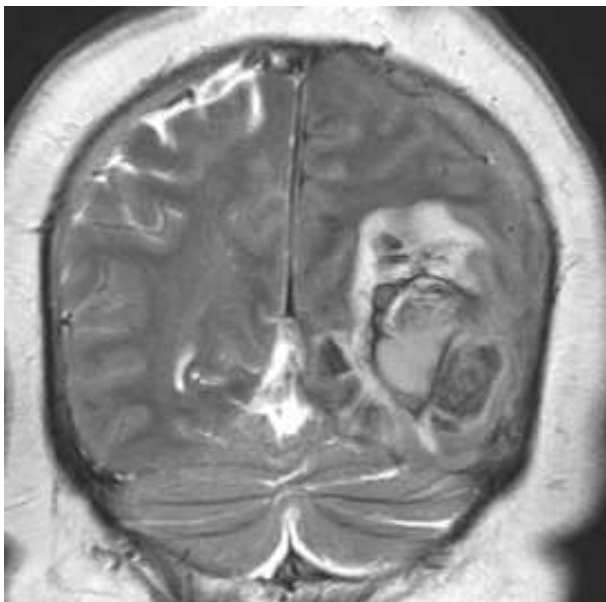


RM-T2



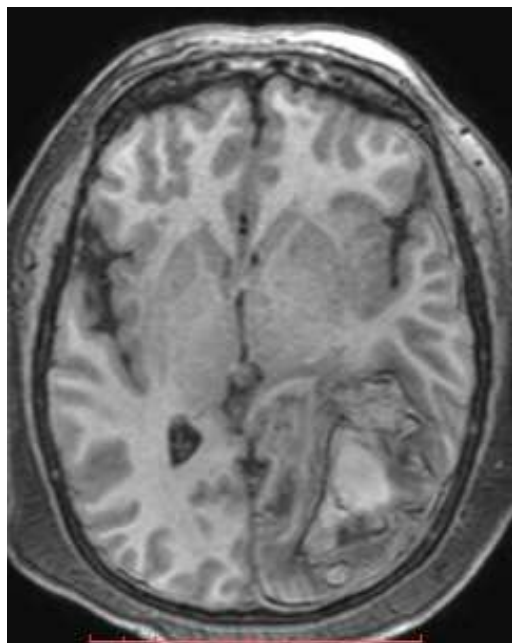
Hemorragia en ganglio basal asociado a HTA

RM-T2

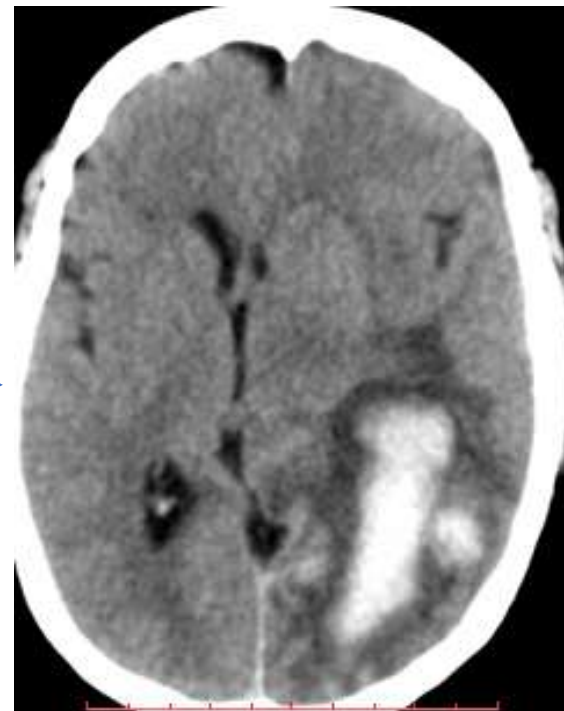


**Mujer de 58
años
En solicitud:
figura
ECV ?**

RM-T1



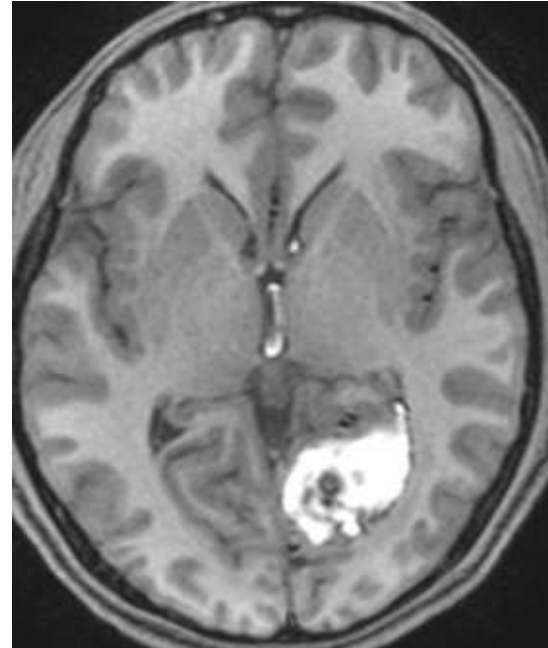
Después de 9 días



Varón de 18 años con cefalea intensa y trastorno de conciencia



Después de 2 días



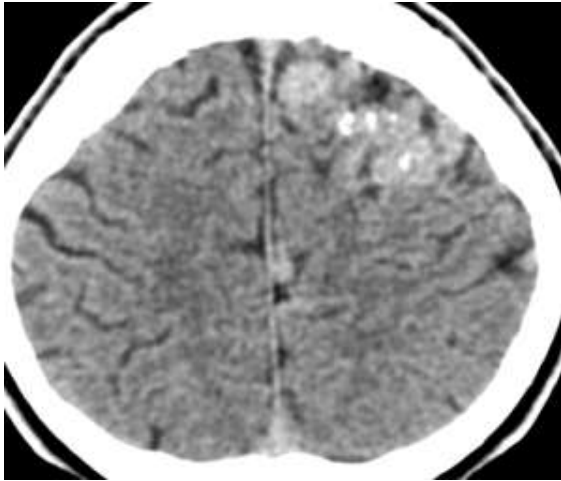
RM-T1 sin
contraste

**Hemorragia intraventricular
y parénquimal**

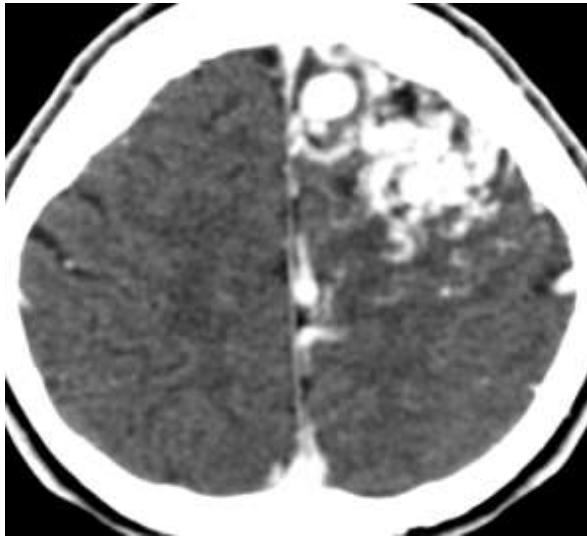
Características por TC y RM de la hemorragia

Estadio	Fisiopatología	TC	RM
Hiperagudo: < 6 h	Oxi hemoglobina intracelular	Hiperdenso	T1 :Isointenso T2 :Hiperintenso
Agudo: 1-3 días	Deoxi hemoglobina intracelular	Hiperdenso	T1 :Iso-hipointenso T2 :Hipointenso
Subagudo Temprano: 4- 7 días.	Meta hemoglobina intracelular Primero en la periferia	Hiperdenso	T1 :Hiperintenso T2: Hipointenso
Subagudo Tardío: 1-4 semanas	Meta hemoglobina extracelular	Isodenso	T1 : Hiperintenso T2 : Hiperintenso
Crónico: meses- años	Hemosiderina y ferritina	Hipodenso	T1 :Hipointenso T2 : Hipointenso

MALFORMACION ARTERIOVENOSA



TC- sin contraste



TC- con contraste

AV Malformation (AVM)



Coronal graphic depicts AVM. Note tightly packed nidus (curved arrow) with intranidal aneurysm (open white arrow), "pedicle" aneurysm (black arrow), and venous varices (open black arrow).

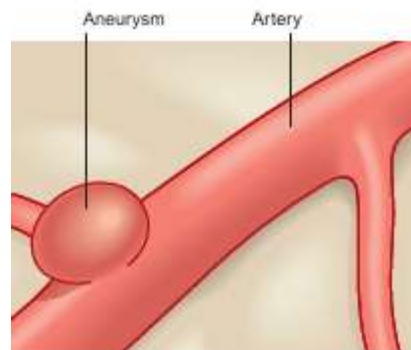


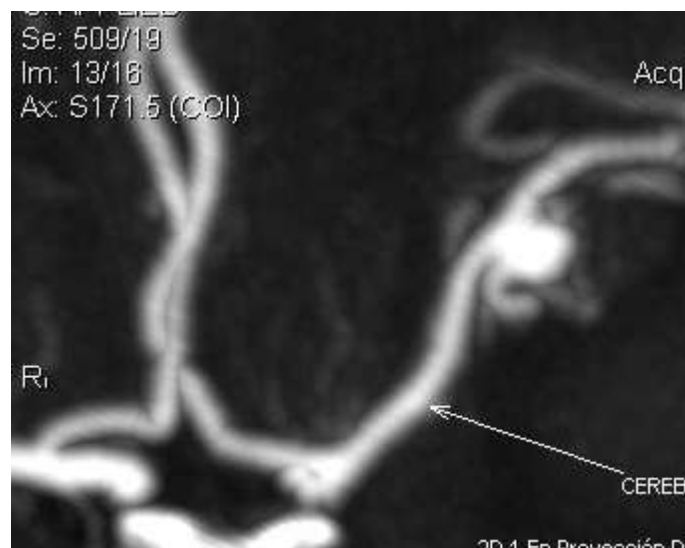
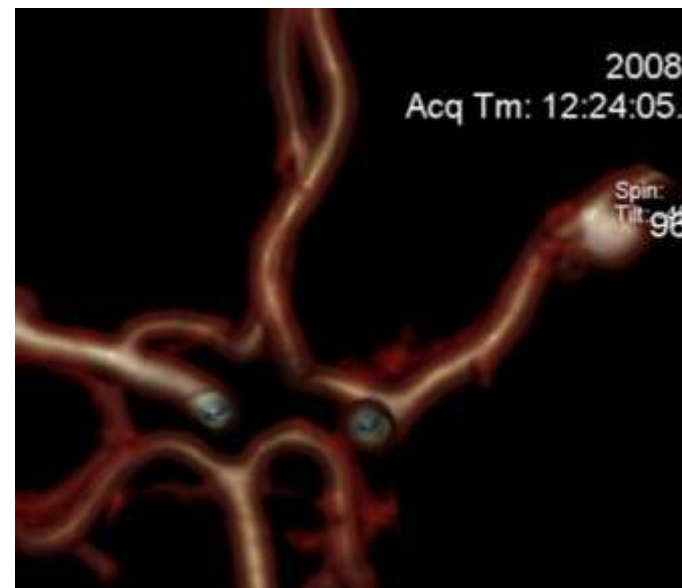
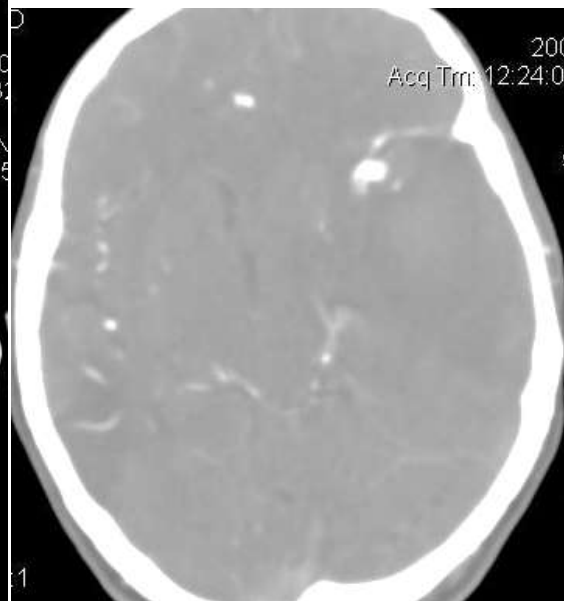
TC- sin contraste



TC- con contraste

ANEURISMA





Aneurisma
Cerebral
media
TCMD

